

VOLOGDA, Russian Federation

WMO: 270370

Lat: 59.3201N

Lon: 39.9250E

Elev: 130

StdP: 99.77

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -29.4	(c) -26.1	(d) -32.4	(e) 0.2	(f) -29.3	(g) -29.1	(h) 0.3	(i) -26.0	(j) 9.5	(k) -6.2	(l) 8.5	(m) -4.9	(n) 1.9	(o) 320	(p) 0.616

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 10.5	(c) 28.5	(d) 19.9	(e) 26.4	(f) 18.8	(g) 24.6	(h) 17.8	(i) 21.1	(j) 26.6	(k) 19.9	(l) 24.8	(m) 18.8	(n) 23.2	(o) 3.2	(p) 150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
19.2	14.1	23.7	18.1	13.2	22.4	17.0	12.3	21.3	61.8	26.7	57.7	25.0	54.0	23.3	26.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8.5	7.4	6.5	-32.8	30.8	3.8	2.0	-35.6	32.3	-37.8	33.4	-39.9	34.6	-42.7	36.0
			-32.9	22.7	3.8	1.8	-35.7	24.0	-37.9	25.0	-40.0	26.0	-42.7	27.3
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	3.5	-10.5	-9.8	-4.3	4.0	10.9	14.9	17.9	15.3	10.0	3.6	-3.0	-7.6		
	DBStd	11.32	8.02	7.93	5.36	4.83	4.97	4.15	3.55	3.64	3.82	4.31	5.96	7.55		
	HDD10.0	3066	636	553	443	191	51	6	1	3	47	202	391	544		
	HDD18.3	5477	894	787	701	431	235	118	52	108	252	457	641	803		
	CDD10.0	701	0	0	0	9	78	154	244	166	45	4	0	0		
	CDD18.3	71	0	0	0	0	4	16	38	13	1	0	0	0		
	CDH23.3	686	0	0	0	3	48	173	341	118	3	0	0	0		
	CDH26.7	159	0	0	0	0	7	40	85	27	0	0	0	0		
(14) Wind	WSAvg	3.2	3.3	3.4	3.4	3.3	3.6	3.1	2.7	2.8	3.0	3.3	3.3	3.4		
(15) Precipitation	PrecAvg	561	34	26	26	33	44	64	74	71	55	50	43	40		
	PrecMax	712	66	55	68	69	97	181	165	175	104	94	82	70		
	PrecMin	360	6	1	3	2	12	9	9	0	8	0	8	16		
	PrecStd	72	13	12	14	17	22	33	40	36	23	22	17	12		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	2.2	3.0	9.7	22.5	27.8	30.1	31.3	29.8	23.5	17.2	8.4	5.5		
		MCWB	1.3	1.6	4.9	13.2	17.9	20.5	21.5	21.8	16.7	12.1	7.5	4.9		
	2%	DB	1.2	1.8	5.9	18.0	24.5	27.8	29.0	26.2	20.6	13.0	6.4	3.6		
		MCWB	0.6	0.8	2.5	10.2	15.7	19.2	20.4	18.8	15.5	10.5	5.5	2.8		
	5%	DB	0.7	1.0	3.9	14.9	22.0	25.2	27.0	24.1	18.1	11.1	5.0	1.8		
		MCWB	0.2	0.3	1.5	8.7	14.1	17.7	19.6	18.0	14.2	9.3	4.3	1.0		
	10%	DB	-0.1	0.2	2.5	11.9	19.8	22.9	25.0	22.1	16.1	9.6	3.6	0.9		
		MCWB	-0.6	-0.5	0.8	6.9	12.7	16.7	18.8	16.9	13.2	8.3	3.0	0.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.5	1.9	5.3	13.2	19.5	21.9	22.8	22.4	18.0	13.2	7.8	5.0		
		MCDB	1.8	2.5	8.7	21.8	25.2	27.8	28.9	28.2	22.0	15.9	8.0	5.4		
	2%	WB	0.7	1.0	3.1	10.8	17.0	20.3	21.5	20.0	16.1	11.0	5.6	2.8		
		MCDB	1.1	1.5	4.9	16.9	22.6	26.0	27.0	24.5	19.1	12.6	6.1	3.4		
	5%	WB	0.2	0.4	1.9	9.1	15.3	18.8	20.4	18.7	14.8	9.6	4.3	1.2		
		MCDB	0.6	0.9	3.3	13.5	20.0	23.7	25.4	22.7	17.3	10.8	4.9	1.6		
	10%	WB	-0.7	-0.5	1.0	7.4	13.7	17.3	19.4	17.5	13.5	8.2	3.0	0.3		
		MCDB	-0.2	0.3	2.1	11.3	18.5	21.6	24.0	21.2	16.0	9.5	3.5	0.8		
(35) Mean Daily Temperature Range	MDBR		6.4	7.4	9.0	9.8	10.9	10.5	10.5	9.8	8.2	5.3	4.3	5.2		
	5% DB	MCDBR	4.6	5.3	9.8	14.4	14.3	13.8	13.4	13.0	11.3	8.2	4.8	3.5		
		MCWBR	4.6	4.9	7.0	8.1	6.9	6.6	6.1	6.4	6.5	6.0	4.5	3.5		
	5% WB	MCDBR	4.4	4.8	7.4	13.1	11.9	12.3	11.9	11.4	9.7	7.6	4.5	3.6		
		MCWBR	4.5	4.7	5.5	7.9	6.5	6.5	5.9	6.1	6.1	5.8	4.4	3.7		
(40) Clear-Sky Solar Irradiance	taub		0.268	0.308	0.334	0.381	0.362	0.360	0.393	0.388	0.353	0.332	0.294	0.251		
	taud		2.061	2.063	2.089	2.185	2.407	2.430	2.357	2.349	2.430	2.366	2.164	2.026		
	Ebn at Noon		505	673	782	815	868	874	830	804	777	667	474	381		
	Edh at Noon		49	84	111	121	105	104	110	103	81	63	45	35		
(44) All-Sky Solar Radiation	RadAvg		0.36	1.08	2.48	3.87	5.19	5.47	5.37	4.06	2.35	1.01	0.33	0.19		
	RadStd		0.06	0.22	0.32	0.44	0.34	0.37	0.60	0.27	0.26	0.16	0.06	0.03		

Historical Trends

		Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (6 neighbors)		+0.69	N/A	N/A	N/A	N/A	N/A	-179	-220	N/A	N/A

Nomenclature: See separate page